

Rare mutations limit of a steady state dispersal evolution model



Selected topics on reaction-diffusion-advection models from spatial ecology

KY Lam, S Liu, Y Lou - arXiv preprint arXiv:2004.07978, 2020 - arxiv.org

We discuss the effects of movement and spatial heterogeneity on population dynamics via reaction-diffusion-advection models, focusing on the persistence, competition, and evolution ...

☆ 36 10

Ideal free dispersal under general spatial heterogeneity and time periodicity

RS Cantrell, C Cosner, KY Lam - SIAM journal on applied mathematics, 2021 - SIAM

A population is said to have an ideal free distribution in a spatially heterogeneous but temporally constant environment if each of its members has chosen a fixed spatial location ...

☆ 28 8

A Hamilton-Jacobi approach to evolution of dispersal

KY Lam, Y Lou, B Perthame - Communications in Partial Differential ..., 2023 - Taylor & Francis

The evolution of dispersal is a classical question in evolutionary biology, and it has been studied in a wide range of mathematical models. A selection-mutation model, in which the ...

☆ 15 11

Adaptation of a quantitative trait to a changing environment: new analytical insights on the asexual and infinitesimal sexual models

J Garnier, O Cotto, E Bouin, T Bourgeron... - Theoretical Population ..., 2023 - Elsevier

Predicting the adaptation of populations to a changing environment is crucial to assess the impact of human activities on biodiversity. Many theoretical studies have tackled this issue ...

☆ 16 16

Dynamics of concentration in a population model structured by age and a phenotypical trait

S Nordmann, B Perthame, C Taing - Acta Applicandae Mathematicae, 2018 - Springer

We study a mathematical model describing the growth process of a population structured by age and a phenotypical trait, subject to aging, competition between individuals and rare ...

☆ 27 12

Influence of a mortality trade-off on the spreading rate of cane toads fronts

E Bouin, MH Chan, C Henderson... - Communications in Partial ..., 2018 - Taylor & Francis

We study the influence of a mortality trade-off in a nonlocal reaction-diffusion-mutation equation that we introduce to model the invasion of cane toads in Australia. This model is ...

☆ 17 7

Stability of Dirac concentrations in an integro-PDE model for evolution of dispersal

KY Lam - Calculus of Variations and Partial Differential ..., 2017 - Springer

We consider an integro-PDE model from evolutionary biology. The solution $u \in (x, \alpha)$ is structured by two variables $x \in D \subset \mathbb{R}^k$ and $\alpha \in (\alpha_-, \alpha_+) \subset \mathbb{R}^+$. The diffusion coefficient in ...

☆ 15 8

Persistence, competition, and evolution

KY Lam, Y Lou - The dynamics of biological systems, 2019 - Springer

In this chapter we discuss some reaction-diffusion models for single and multiple populations in spatially heterogeneous environments and advective environments. Our goal ...

☆ 12 5

Concentration phenomena in an integro-PDE model for evolution of conditional dispersal

W Hao, KY Lam, Y Lou - Indiana University Mathematics Journal, 2019 - JSTOR

In order to study the evolution of conditional dispersal, we extend the Perthame-Souganidis mutation-selection model and consider an integro-PDE model for a population structured by ...

☆  13 8

Dirac-concentrations in an integro-PDE model from evolutionary game theory

KY Lam - Discrete & Continuous Dynamical Systems-B, 2017 - par.nsf.gov

Nonlocal Lotka-Volterra models have the property that solutions concentrate as Dirac masses in the limit of small diffusion. Motivated by the existence of moving Dirac ...

☆  8 7 

PDF



AI



